

Exam Score Frequency Distribution — Line-by-Line Solution

Easy method for calculating Bin, Frequency, Cumulative Frequency and Cumulative %

1. Given Exam Scores

42, 45, 47, 49, 50, 52, 53, 55, 56, 58, 60, 61, 62, 64, 65, 67, 68, 70, 72, 74, 75, 76, 78, 80, 82, 84, 85, 87, 88, 90, 92, 94, 95, 97, 99, 100, 102, 104, 106, 108

Total number of scores = **40**

2. What does a Bin mean?

A bin is the upper limit of a group. For example, Bin 53 means: count scores greater than 42 and up to 53.

3. Calculate Frequency — Bin 42

Scores ≤ 42 : **42**

Count = **1** → Frequency = **1**

4. Calculate Frequency — Bin 53

Scores from 43 to 53: **45, 47, 49, 50, 52, 53**

Count = **6** → Frequency = **6**

5. Calculate Frequency — Bin 64

Scores from 54 to 64: **55, 56, 58, 60, 61, 62, 64**

Count = **7** → Frequency = **7**

6. Calculate Frequency — Bin 75

Scores from 65 to 75: **65, 67, 68, 70, 72, 74, 75**

Count = **7** → Frequency = **7**

7. Calculate Frequency — Bin 86

Scores from 76 to 86: **76, 78, 80, 82, 84, 85**

Count = **6** → Frequency = **6**

8. Calculate Frequency — Bin 97

Scores from 87 to 97: **87, 88, 90, 92, 94, 95, 97**

Count = **7** → Frequency = **7**

9. Calculate Frequency — More

Scores greater than 97: **99, 100, 102, 104, 106, 108**

Count = **6** → Frequency = **6**

10. Cumulative Frequency

Easy rule: **keep adding the frequencies from top to bottom.**

Bin 42: $1 = 1$

Bin 53: $1 + 6 = 7$

Bin 64: $7 + 7 = 14$

Bin 75: $14 + 7 = 21$

Bin 86: $21 + 6 = 27$

Bin 97: $27 + 7 = 34$

More: $34 + 6 = 40$

11. Cumulative Percentage

Formula: **Cumulative % = (Cumulative Frequency ÷ Total Frequency) × 100**

Total Frequency = **40**

Bin 42: $(1 \div 40) \times 100 = 2.50\%$

Bin 53: $(7 \div 40) \times 100 = 17.50\%$

Bin 64: $(14 \div 40) \times 100 = 35.00\%$

Bin 75: $(21 \div 40) \times 100 = 52.50\%$

Bin 86: $(27 \div 40) \times 100 = 67.50\%$

Bin 97: $(34 \div 40) \times 100 = 85.00\%$

More: $(40 \div 40) \times 100 = 100.00\%$

12. Final Answer

Bin	Frequency	Cumulative Frequency	Cumulative %
42	1	1	2.50%
53	6	7	17.50%
64	7	14	35.00%
75	7	21	52.50%
86	6	27	67.50%
97	7	34	85.00%
More	6	40	100.00%

Remember: Frequency = count values in each range. Cumulative Frequency = keep adding frequencies. Cumulative % = $(\text{Cumulative Frequency} \div 40) \times 100$.